

Lecture 3 Gauss's law

Big ideas for today

- How to find $\vec{E}$ outside sphere with uniform ρ
- Introducing flux
 - how much vector field passes through an area
- Find flux of $\vec{E}$ through a sphere surrounding a point charge
- Gauss's law:
 - Always true, sometimes useful for finding $\vec{E}$.

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

Gauss's law

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

If we have a closed surface, (i.e. a bubble that fully encloses a volume), the flux Φ_E of $\vec{E}$ out through the closed surface is equal to the charge enclosed divided by ϵ_0

We can use gauss's law ONLY if we have cylindrical, spherical, or planar symmetry around an object. We use that to reduce the number of unknowns, then pull out the $\vec{E}$ from the surface integral and solve.

$$(\Phi_c) = \vec{c} \cdot d\vec{A}$$

if $\vec{c}$ is constant and the area is flat, then $\Phi = \vec{c} \cdot \vec{A}$

Lets do a quick example. Imagine water flowing through a pipe. The flux is

$$\begin{aligned} \Phi &= \vec{v} \cdot \vec{A} = vA \cos \theta \\ \end{aligned}$$

We can cut a different surface for the pipe. Let's try a more electric – y example.

$$\begin{aligned} \vec{E} &= \frac{\lambda}{2\pi \epsilon_0 r} \hat{r} \\ \end{aligned}$$

Lets use cylindrical coordinates (r, θ, z) . We have shown a fast way of finding the field using gauss's law.